



Evaluation of preoperative intraperitoneal injection of bupivacaine for intraoperative antinociception in cats undergoing ovariohysterectomy: A randomized clinical trial

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ABSTRACT

Objective: To determine if preoperative intraperitoneal bupivacaine can minimize intra- and postoperative nociception/pain in cats undergoing ovariohysterectomy.

Study design: Prospective, randomized, investigator-blinded, placebo-controlled clinical trial.

Animals: Forty-seven, intact female cats.

Methods: Cats were anesthetized using a standard protocol and randomized to receive ultrasound-guided intraperitoneal 0.9 % saline (US-S) or 0.25 % bupivacaine (US-IPLA) before ovariohysterectomy. On recovery, US-S cats received 2 mg/kg robenacoxib subcutaneously and US-IPLA cats received equivalent volume of 0.9 % saline subcutaneously. Intraoperative outcome variables included heart rate (HR), respiratory rate (fR), systolic arterial pressure (SAP), and vaporizer setting associated with relevant surgical events characterized by manipulation of each ovarian pedicle (OP1, OP2) and the uterine body (UB). The postoperative outcome variable was need for rescue analgesia, determined using the Glasgow Composite Measure Pain Scale during 6 h after tracheal extubation. Intraoperative data were analyzed using two-way ANOVA and Sidák's multiple comparisons test. The probability of postoperative rescue analgesia was analyzed using the Gehan-Breslow-Wilcoxon test. Significance was $p < 0.05$.

Results: Compared to baseline, all surgical events caused significant increases in HR and SAP in both groups, fR increased in US-IPLA but not in US-S, vaporizer settings remained unchanged during OP1, OP2 and UB in group US-IPLA, and were significantly higher only during OP1 in group US-S. There were no significant between-group differences in intraoperative variables and postoperative need for rescue analgesia.

Conclusions and clinical relevance: Preoperative intraperitoneal bupivacaine had minimal effects on intraoperative indicators of nociception. The need for rescue analgesia was not significantly different between groups.

1. Introduction

Effective pain management is essential for patient care as it minimizes animal suffering, reduces anesthetic requirements, and allows for comfortable postoperative recovery. In humans, the presence of pain during or after surgery has been identified as a predictor of chronic pain after abdominal surgery (Brandsborg et al., 2008; Macrae, 2008; Montes et al., 2015; Gan, 2017). This phenomenon is likely underappreciated in animals due to the difficulties in pain assessment and follow-up.

Preemptive analgesia aims to prevent the development of central

sensitization by administering analgesic medications before surgical trauma (Katz et al., 1992; Kissin, 2000). Local anesthetic techniques are a critical component of multimodal analgesia and, when effective, provide complete blockade of nociceptive transmission from the periphery to the spinal cord. Ultrasound has revolutionized the way local blocks are performed, allowing for real-time visualization of local anesthetic injection at a given target, improving efficacy and safety (Mariano et al., 2014). Techniques such as the transversus abdominis plane and rectus sheath blocks offer somatic analgesia for abdominal procedures in cats using a midline surgical approach (Garbin et al., 2023; Haley et al.,

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2024; Josso et al., 2022); however, support for visceral analgesia, especially intraoperatively, is lacking. Intraperitoneal local anesthetic administration (IPLA) is a technique previously used intraoperatively in cats to provide visceral analgesia for ovariohysterectomy (OHE). When used as a “splash block” in the abdomen just before OHE, the analgesic effects of intraoperative IPLA were demonstrated in the postoperative period (Benito et al., 2016). Minimal time between application of a splash block with bupivacaine and beginning OHE could limit the efficacy of intraoperative visceral analgesia. Studies investigating ovarian pedicle blocks where lidocaine was administered directly into the pedicle after exteriorization have demonstrated a reduction in intraoperative ketamine boluses required for OHE in cats suggesting an analgesic benefit (Zilberstein et al., 2008). Bupivacaine has a longer duration of effect, but also a potentially longer time to onset compared to lidocaine (Grubb and Lobprise, 2020). Preoperative ultrasound-guided IPLA offers a novel technique that could allow adequate onset time for bupivacaine to provide intraoperative nociceptive control in addition to the postoperative benefits, thus extending the visceral analgesic benefits of IPLA.

When ultrasound-guided rectus sheath block (US-RSB) with bupivacaine was compared to robenacoxib for intraoperative and postoperative analgesia in cats undergoing OHE, US-RSB had possible benefits on isoflurane requirement during celiotomy and closure of body wall, but markedly worse postoperative pain control as compared to robenacoxib (Haley et al., 2024). These findings suggest that postoperative pain following ovariohysterectomy in cats may stem from a marked visceral component.

The primary hypothesis tested in the current study was that preoperative ultrasound-guided IPLA with bupivacaine would produce significantly greater intraoperative antinociception compared to saline control while investigating if the postoperative benefits seen in other studies would be maintained. The purpose of this study was to develop a novel preemptive analgesic technique for ovariohysterectomy in cats.

2. Materials and methods

2.1. Study design

This study was designed as a prospective, randomized, blinded, and controlled clinical trial. All enrolled cats were female, domestic, and considered healthy based on physical examination, packed cell volume (PCV), and total protein (TP). All cats were enrolled between October and November 2022 as part of a third-year veterinary student spay and neuter teaching laboratory and were obtained from local shelters. Cats were excluded if, on initial physical exam, they were deemed by the clinician performing pain assessment to be too aggressive for repeated handling. Pregnant cats were not excluded. Written, informed consent was obtained from the owners of all cats. The study adhered to guidelines for the ethics and welfare of animals and was approved by the Institutional Animal Care and Use Committee of the University of Minnesota protocol number 2108-39358 A. Cats were randomly assigned with simple randomization by an investigator not involved in data collection or treatment administration using a random number generator (www.randomization.com) to receive ultrasound-guided intraperitoneal injections, as described below, of either 0.4 ml/kg per hemiabdomen 0.9 % saline (US-S; Hospira Inc., IL, USA) or 0.4 ml/kg per hemiabdomen 0.25 % bupivacaine (US-IPLA; Bupivacaine Hydrochloride Inj, 5 mg/ml; Hospira Inc., IL, USA). The assigned solution for each cat was drawn into sterile syringes labeled “US” by an investigator who was not involved in data collection or treatment administration. At the end of surgery, the US-S group received robenacoxib (2 mg/kg subcutaneous (SC); Onsior, 20 mg/ml; Elanco Animal Health Inc., IN, USA) and the US-IPLA group received an equal volume of 0.9 % saline SC. The recovery solution was drawn into sterile syringes labeled “Recovery” by an investigator not involved in data collection or treatment administration. Recovery solution was administered with a 20 g needle to

maintain blinding in the face of differing solution viscosity.

2.2. Anesthetic protocol, monitoring and support

All cats were premedicated with buprenorphine (0.24 mg/kg SC; Simbadol, 1.8 mg/ml; Zoetis Inc., NJ, USA), dexmedetomidine (0.005 mg/kg intramuscular (IM); Dexdomitor, 0.5 mg/ml; Zoetis Inc., NJ, USA), and ketamine (5 mg/kg IM; Zetamine, 100 mg/ml; VetOne, ID, USA). An intravenous (IV) catheter was placed once sedation was evident. The abdomen was clipped, and then the cats were moved into the surgical theater. General anesthesia was induced with propofol (1–6 mg/kg intravenous to effect; Propofol Injectable Emulsion USP, 10 mg/ml; Sagent Pharmaceuticals Inc., IL, USA), and 0.1–0.2 ml of 2 % lidocaine was sprayed topically on the arytenoids for desensitization. Orotracheal intubation was performed with an appropriately sized cuffed endotracheal tube. Anesthesia was maintained with isoflurane delivered in oxygen with a Mapleson D system, using a vaporizer whose output calibration was verified by a third-party vendor before beginning the study. Oxygen flow rate was set at 200 ml/kg/min, and end-tidal carbon dioxide was monitored with a mainstream portable capnograph to ensure that oxygen flow remained higher than minute volume evidenced by no rebreathing of CO₂ (EMMA capnograph; Masimo Corporation, CA, USA). Heart rate (HR) measured by counting the auditory pulse rate from a doppler, respiratory rate (fR) measured with the mainstream portable capnograph and verified visually by monitoring chest excursions, body temperature measured in the oropharynx with a digital thermometer, oxygen hemoglobin saturation (SpO₂) obtained with a probe on the tongue (Pulse Oximeter 2500 A Vet, Nonin Medical Inc., MN, USA), systolic arterial pressure (SAP) measured via Doppler technique (Doppler Flow Detector; Parks Medical Electronics Inc., OR, USA) with a crystal placed on the palmar surface and a blood pressure cuff proximal to the carpus on a thoracic limb (sized to approximately 40 % limb circumference), vaporizer setting, and anesthetic depth were continuously monitored and recorded every 5 min. Lactated Ringer's solution (3 ml/kg/h IV; Baxter Healthcare Corporation, IL, USA) was delivered throughout surgery with a 60 drop/ml infusion set which was monitored for appropriate drip rate every 15 min.

Anesthetic depth was adjusted by student anesthetists under the direction of anesthesia instructors to maintain minimal jaw tone, ventromedial rotation of the eyes, and no palpebral reflex. Movement was treated with propofol (1 mg/kg IV) followed by increasing the vaporizer setting by 0.25 %. Hypotension was treated based on anesthesia instructor discretion and patient presentation by decreasing vaporizer setting if anesthetic depth allowed, with atropine (0.02 mg/kg IV, Atropine Sulfate Injection USP, 0.4 mg/ml; Intas Pharmaceuticals Limited, Gujarat, India) if cats were determined to be bradycardic, dose repeated as necessary, ephedrine (0.1 mg/kg IV, Ephedrine Sulfate Injection USP, 50 mg/ml; Lifestar Pharma LLC, NJ, USA) if cats were euhydrated and not bradycardic, or Lactated Ringer's solution bolus (10 ml/kg) to assess volume responsiveness, all with the goal of maintaining SAP $\geq$ 90 mmHg. Blood pressure was stabilized prior to the beginning of surgery and baseline values were taken after administration of blood pressure support. Heat support was provided using a forced warm air device (BedJet, RI, USA) placed underneath the cat.

Surgery was performed by third-year veterinary students with oversight from surgery faculty or residents. Each spay was performed by a different pair of students.

2.3. Ultrasound-guided intraperitoneal injections

As a pilot study, the ultrasound technique (described in further detail below) was tested in two fresh cat cadavers by injecting 0.8 ml/kg 74 % ioversol contrast (Optiray 350; Guebert LLC, NJ, USA), and a computed tomography scan was performed which showed deposition of the contrast in the region caudal to the kidneys in the dorsal abdominal cavity.

For the present study - with each cat in dorsal recumbency, the skin over the mid-abdomen was aseptically prepared using chlorhexidine and alcohol scrub solutions. Wearing sterile gloves, the intraperitoneal injections were performed by placing a 6–13 Mhz linear ultrasound probe (L25 Probe; Sonosite Inc., WA, USA), which was cleaned with alcohol between uses, on the abdomen in transverse orientation lateral to the umbilicus using alcohol as a contact medium. The caudal pole of the kidney was visualized to aid orientation and injection near the predicted anatomic location of the ovary. The ovary was not visualized before injection. The probe was variably moved caudal to where the kidney was visualized where a window could be found that would avoid puncture of abdominal organs and a 22 gauge, 1.5 in. (25 mm) spinal needle (Quincke Spinal Needle; Becton, Dickinson and Company, NJ, USA) connected directly to a syringe containing the assigned solution was advanced in a medial to lateral direction using an in-plane technique. The needle was advanced through the external oblique, internal oblique, and transversus abdominis muscles before penetrating the peritoneum under US visualization. Following negative aspiration, half of the syringe contents were injected into the peritoneal cavity. The same procedure was repeated on the contralateral side. Following treatment injection, the surgical field was aseptically prepared with alternating alcohol and chlorhexidine scrubs. Cats remained in dorsal recumbency, and a ventral midline approach was used for OHE.

2.4. Intraoperative outcome variables

Nociceptive responses to surgical stimuli, inferred from increases in HR, fR or SAP, and anesthetic requirement, inferred from vaporizer settings to maintain an adequate surgical depth of anesthesia, were used as outcome variables. These variables were determined and recorded before skin incision (baseline), and at surgical events expected to be affected by IPLA, including manipulation and transection of each ovarian pedicle (OP1, OP2), and the uterine body (UB). The highest recorded value of HR, fR, SAP and vaporizer setting during each of these surgical events was used for statistical analysis. Data were collected in real time by a veterinary anesthesia instructor blinded to treatment allocation.

2.5. Postoperative outcome variables

The main postoperative outcome variable was a yes/no analysis of the need for rescue analgesia, which was determined using the feline short form Glasgow Composite Measure Pain Scale (GCMPs) at 0.75, 1.5, 3, and 6 h after tracheal extubation. Cats were additionally monitored for anesthetic recovery and pain by trained veterinary technicians or residents for the duration of the postoperative period as part of the normal laboratory guidelines. If pain concerns were identified outside the study time points, the pain would be addressed and treated at the clinician's discretion with this variation being noted. Additionally, if rescue analgesia was required the cats were reassessed 30 min after rescue to ensure adequate pain control. Additional analgesics would be administered at the clinician's discretion if the protocol rescue was inadequate, and this would be noted. As per the GCMPs, rescue analgesia was provided if pain scores were $\geq 5/20$. Rescue analgesia was as follows: cats in the US-S group were given buprenorphine (0.01 mg/kg SC) while those in the US-IPLA group were given robenacoxib (2 mg/kg SC). Buprenorphine was diluted with saline to the equivalent volume of a 2 mg/kg injection of robenacoxib. The assigned solutions were drawn into syringes by an investigator not involved in data collection or treatment administration and labeled "Rescue". After the 6-h study period, all cats were injected with the contents of their assigned rescue syringe if they had not already received it to ensure postoperative analgesics were given to all cats as per the previously mentioned Institutional Animal Care and Use Committee approved laboratory protocol.

2.6. Sample size calculation

The main goal of this study was to determine whether preoperative IP bupivacaine could significantly minimize nociceptive responses to surgical manipulation of ovarian pedicles and uterine body. As such, the sample size was calculated considering the hemodynamic response to surgical stimulation as the main variable. Variables used were obtained from preliminary data in control cats (Haley et al., manuscript accepted) showing a mean $\pm$ SD increase of 64 ± 41 mmHg in SAP during OP manipulation and transection, and assuming that an effect size of an approximately 50 % reduction in SAP change (32 mmHg) would be clinically significant. Using these values, a sample size of 24 cats per group had 80 % power to detect a 33 mmHg change in SAP with a significance level (alpha) of 0.05 (two-tailed) per group. Although comparing the need for rescue analgesia postoperatively was not the main goal of this study, the sample size provided 80 % power to detect a significant difference between groups (chi-square for two proportions, alpha 0.05, two-tailed) if the proportion of success (i.e., no need for rescue analgesia) in group US-S was at least twice that in group US-IPLA group, as shown previously (Benito et al., 2016). Sample size estimation was performed using commercial software (GraphPad StatMate, GraphPad Software, Inc., CA, USA).

2.7. Statistics

Data analysis was conducted using commercially available statistics software (Prism 10 for MacOS, v.10.0.2, GraphPad Software). For all analyses, a significance level of $p < 0.05$ was used. Normality was assessed by visual inspection of QQ plots and the D'Agostino-Pearson omnibus normality test. Data for HR, SAP, fR, and vaporizer settings were normally distributed and analyzed using two-way analysis of variance (ANOVA) followed by Šidák's multiple comparisons test, while correcting for multiple comparisons using statistical hypothesis testing. Postoperative rescue analgesia was compared between the two groups by analyzing the probability of no event (i.e., no need for rescue analgesia or "survival") using the Gehan-Breslow-Wilcoxon test. The odds ratio of the need for rescue analgesia was analyzed using the Mantel-Haenszel test. The proportion of cats in each group requiring rescue analgesia was determined and compared using Fisher's exact test. Data are shown as mean $\pm$ standard deviation (SD) if normally distributed or median and interquartile range (IQR) or range if not normally distributed.

3. Results

3.1. Demographic variables

A total of 47 healthy female cats (US-IPLA = 24, US-S = 23) sourced from local shelters, between approximately 4 months and 6 years old, with body masses ranging between 1.3 and 5.0 kg were included in this study. No cats presented as pregnant. After the US-guided injection, time to skin incision was 21 ± 7 min for US-S and 24 ± 7 min for US-IPLA ($p = 0.194$; Fig. 1), with corresponding time to first pedicle manipulation of 50 ± 15 min and 41 ± 5 min ($p = 0.185$; Fig. 1). Total surgery time was 104 ± 24 min for US-S and 103 ± 26 min for US-IPLA ($p = 0.963$; Fig. 1).

3.2. Intraoperative outcomes

Compared to baseline, there were significant increases in HR and SAP during all evaluated surgical events (OP1, OP2 and UB) in both groups ($p < 0.0001$ for all comparisons; Fig. 2). Compared to baseline, fR increased in the US-IPLA group during all surgical events (OP1, $p = 0.026$; OP2, $p = 0.004$; UB $p = 0.001$; Fig. 2) but not in the US-S group (OP1, $p = 0.902$; OP2, $p = 0.139$; UB $p = 0.078$; Fig. 2). Vaporizer settings were not significantly different from baseline in the US-IPLA group

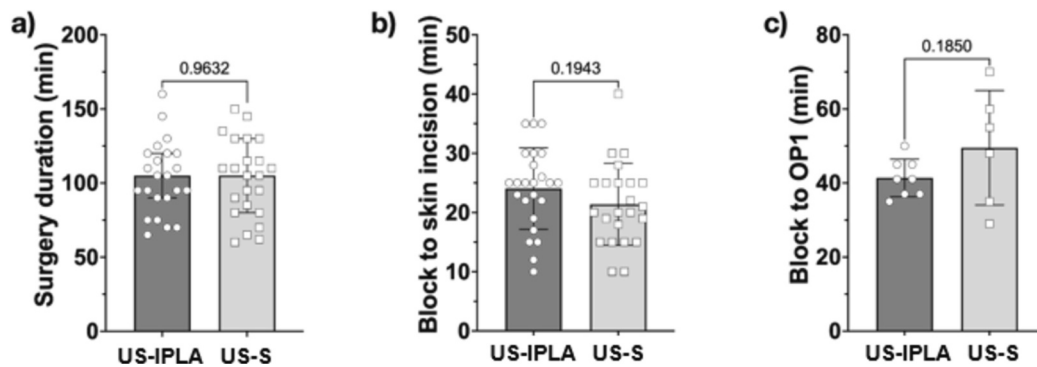


Fig. 1. Mean (bar), standard deviation (whiskers) and individual values (symbols) of the duration of surgery (panel a), time from block to skin incision (panel b) and time from block to manipulation of first ovarian pedicle (OP1, c) in cats undergoing surgery for ovariohysterectomy ($n = 47$). Cats were randomly assigned to be given preoperative ultrasound-guided intraperitoneal injections directed towards the approximate location of each ovary. Injectate consisted of either 0.25 % bupivacaine (US-IPLA; 0.8 ml/kg/site, $n = 24$) or a similar volume of saline/site (US-S; $n = 23$). Group means were not significantly different ($p = 0.963$ for surgery duration; $p = 0.194$ for time from block to skin incision; $p = 0.185$ for time from block to OP1).

Mean ± SD:

Surgery duration: US-IPLA 104 ± 24 min; US-S 103 ± 26 min.

Block to skin incision: US-IPLA 24 ± 7 min; US-S 21 ± 7 min.

Block to OP1: US-IPLA 41 ± 5 min; US-S 50 ± 15 min.

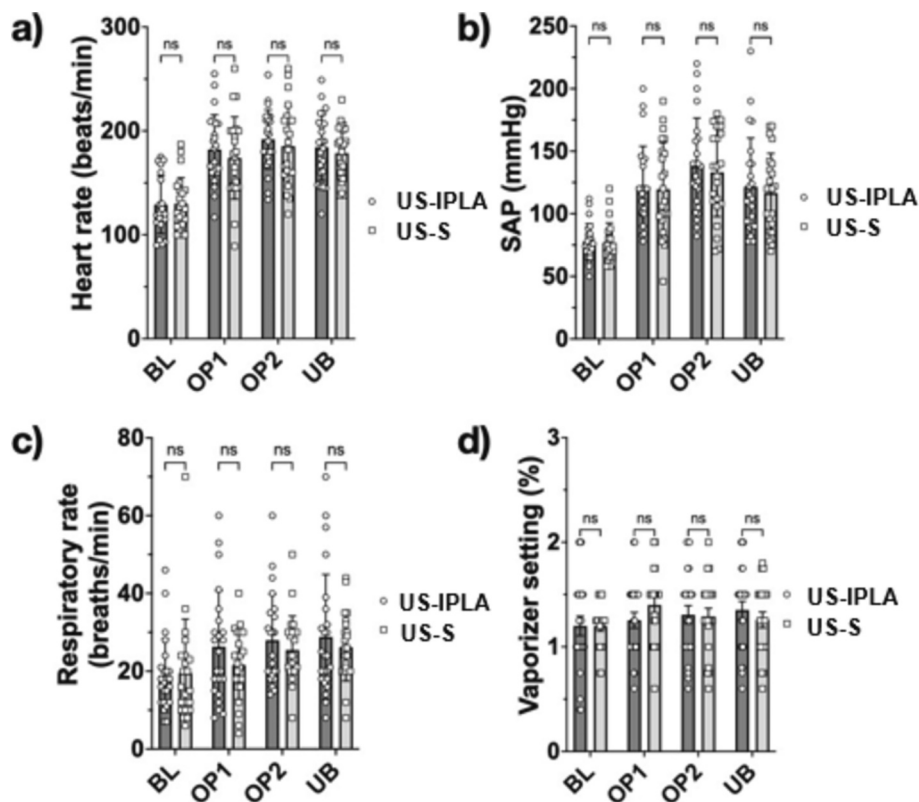


Fig. 2. Mean ± SD (bars and whiskers) and individual values (symbols) of heart rate (panel a), systolic arterial pressure (SAP; panel b), respiratory rate (panel c) and vaporizer settings (panel d) during isoflurane anesthesia and surgery for ovariohysterectomy in cats ($n = 47$). Maximum values for each variable were recorded before skin incision (BL), and during exposure, ligation and excision of first (OP1) and second (OP2) ovarian pedicles, and the uterine body (UB). $p < 0.05$ considered significant; ns = not significant.

for any of the surgical events (OP1, $p = 0.865$; OP2, $p = 0.436$; UB $p = 0.149$; Fig. 2), whereas they were significantly higher than baseline during OP1 ($p = 0.048$) but not during OP2 ($p = 0.604$) or UB ($p = 0.855$) in the US-S group. Between-group comparisons showed no significant differences in HR, SAP, fR, or vaporizer settings during any of the surgical events ($p > 0.05$). No significant complications from performing the intraperitoneal block or from the surgical procedure were

noted in any cat.

The median (IQR) propofol dose administered intraoperatively to help maintain an adequate plane of anesthesia was 1.1 mg/kg (0, 2.3) in US-S and 1.3 mg/kg (0, 1.9) in US-IPLA ($p = 0.907$) while the proportion of cats requiring additional propofol was 14/23 in US-S and 15/24 in US-IPLA ($p > 0.999$; Fig. 3). The proportion of cats treated with atropine was 20/23 in the US-S and 13/24 in the US-IPLA and was significantly

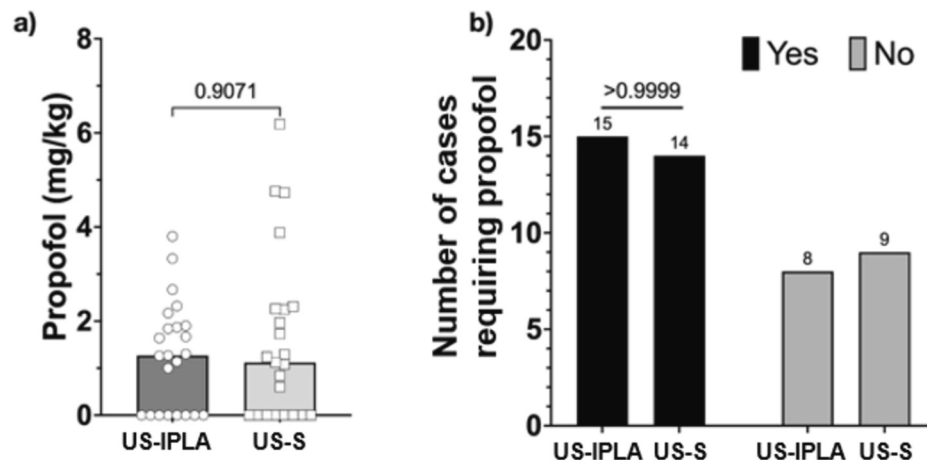


Fig. 3. Median (bar), interquartile range (whiskers) and individual values (symbols) of the amount of propofol used intraoperatively (panel a) and the proportion of cats receiving propofol intraoperatively (panel b) in cats undergoing surgery for ovariohysterectomy ($n = 47$). Cats were randomly assigned to be given preoperative ultrasound-guided intraperitoneal injections directed towards the approximate location of each ovary. Injectate consisted of either 0.25 % bupivacaine (US-IPLA; 0.8 ml/kg/site, $n = 24$) or a similar volume of saline/site (US-S; $n = 23$). Groups were not significantly different ($p = 0.907$). The proportion of cats administered propofol was not significantly different between groups ($p > 0.999$).

Median and IQR:

US-IPLA 1.3 mg/kg (0, 1.9), US-S: 1.1 mg/kg (0, 2.3).

different between groups ($p = 0.047$; Fig. 4). The proportion of cats requiring ephedrine (7/23 in US-S and 6/24 in US-IPLA; $p > 0.999$) and fluid boluses (5/23 in US-S and 2/24 in US-IPLA; $p = 0.245$; Fig. 4) were not significantly different between groups.

3.3. Postoperative outcomes

No cats required treatment outside of the study time points and all cats that required rescue analgesia showed appropriate improvement in their pain scores with rescue treatment. No significant differences were found between the two groups for the need for rescue analgesia, as assessed by the probability of no event (i.e., no need for rescue analgesia; $p = 0.443$; Fig. 5). Similarly, the proportion of cats requiring rescue analgesia within 6 h after tracheal extubation (10/24 US-IPLA, 7/23 US-S) was not significantly different between the two groups ($p = 0.547$; Fig. 5). The odds ratio indicated that cats in the US-IPLA group were 1.49 times (95 % CI 0.55, 4.03) more likely to require rescue analgesia than cats in the US-S group but this difference was not statistically significant ($p = 0.547$; Fig. 5).

4. Discussion

The primary goal of the present study was to test the hypothesis that preoperative intraperitoneal administration of bupivacaine via ultrasound guidance would produce significant intraoperative antinociception compared to saline control using hemodynamic changes as an indicator for nociception. The results did not support the primary hypothesis because the change in intraoperative outcome variables associated with the relevant surgical events was not significantly different between the two groups. Within-group comparisons suggested that intraperitoneal bupivacaine may have had some sparing effect on isoflurane requirement, although the difference was unlikely to be clinically significant.

There were significant increases in SAP during surgical manipulation of both OP and UB that were similarly marked in both groups, indicating that the lack of significant difference between groups was not due to a preemptive antinociceptive effect of pre-anesthetic medications masking hemodynamic responses. A type II error is also unlikely because the study was adequately powered to detect what was deemed to be a clinically significant mitigation in autonomic response to nociception, as

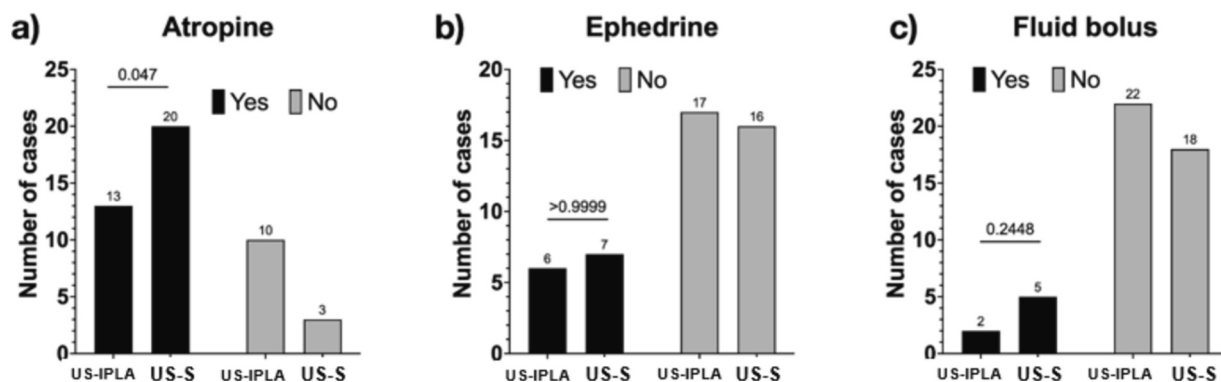


Fig. 4. Proportion of cats requiring and not requiring atropine (panel a), ephedrine (panel b) and fluid boluses (panel c) in cats submitted to ovariohysterectomy ($n = 47$). Cats were randomly assigned to be given preoperative ultrasound-guided intraperitoneal injections directed towards the approximate location of each ovary. Injectate consisted of either 0.25 % bupivacaine (US-IPLA; 0.8 ml/kg/site, $n = 24$) or a similar volume of saline/site (US-S; $n = 23$). The proportion of cats given atropine was significantly different between groups ($n = 13/24$ cats in Group US-IPLA and $n = 20/23$ cats in Group US-S; $p = 0.047$), but not the proportion of cats administered ephedrine ($n = 6/24$ cats in Group US-IPLA and $n = 7/23$ cats in Group US-S; $p > 0.999$) or fluid boluses ($n = 2/24$ cats in Group US-IPLA and $n = 5/23$ cats in Group US-S; $p = 0.245$).

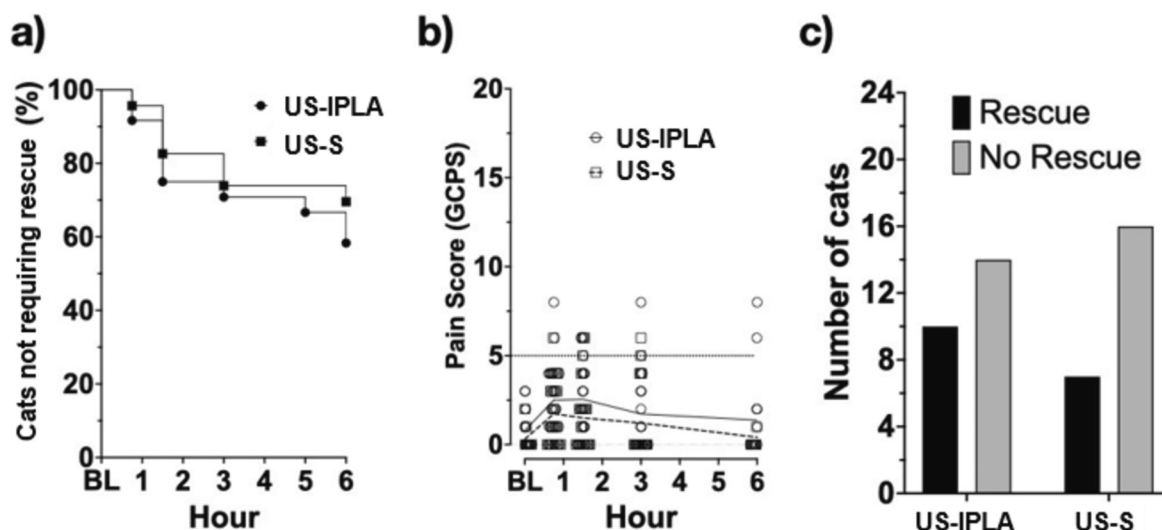


Fig. 5. Probability of no event (i.e., no need for rescue analgesia; panel a), median and interquartile range of pain scores (panel b) and proportion of cats requiring and not requiring rescue analgesia (panel c) in cats submitted to ovariohysterectomy ($n = 47$). A single investigator unaware of treatment allocation assessed pain preoperatively (i.e., baseline or BL) and at 0.75, 1.5, 3 and 6 h postoperatively using the Glasgow Feline Composite Measure Pain Scale (GCMPS). Rescue analgesia for cats with a pain score ≥ 5 (dotted line) was provided with robenacoxib (2 mg/kg subcutaneously; US-IPLA group) or buprenorphine (0.01 mg/kg subcutaneously; US-S group). In panel b, pain scores after rescue analgesia ($n = 10$ cats in US-IPLA and $n = 7$ cats in US-S) were excluded from calculations. The probability of no event was not significantly different between groups ($p = 0.443$). Cats in US-IPLA were 1.49 times (95 % CI 0.55, 4.03) more likely to require rescue analgesia than cats in US-S. In panel c, the proportion of cats requiring rescue analgesia ($n = 10/24$ cats in US-IPLA and $n = 7/23$ cats in US-S) was not significantly different between groups ($p = 0.547$).

indicated by SAP responses during OP manipulation. In addition, the mean difference in SAP between the two groups during surgical manipulation of OP and UB was in the single digits, further strengthening the assertion that the ultrasound-guided intraperitoneal block with bupivacaine did not have a major antinociceptive effect on noxious stimuli intraoperatively. It is unlikely that the volume of injectate did not adequately distribute ventrally to the level of the ovarian pedicles based on the preliminary results in the 2 cat cadavers (4 hemiabdomens), which consistently showed the presence of contrast pooling in the region where the ovarian pedicles are anatomically located. The possibility that the concentration of bupivacaine or volume of injectate was insufficient for complete nerve blockade cannot be ruled out and further studies are required to test this notion. However, some relatively minor effects on the overall anesthetic management might have occurred. For example, the significantly greater need for atropine in the US-S group could be due to the higher vaporizer setting and more cardiovascular depression compared to the baseline in this group. The lack of increase in respiratory rate in the US-S group during OP and UB manipulation, in contrast to the increases observed in the US-IPLA group, could also be related to a deeper anesthetic depth in the US-S compared to the US-IPLA. Thus, the within-group differences suggest that there may be some contribution to decreasing isoflurane requirements, which may have reflected in better cardiopulmonary function during anesthesia.

The breakdown of the suspensory ligament and exteriorization of the ovary is one of the most nociceptive parts of ovariohysterectomy surgery. Priority was given to blocking this area while also investigating if the generalized intraperitoneal approach would desensitize additional structures such as the uterine body. In addition, the landmarks for this area are easier to ascertain on ultrasound and with the ultimate goal of developing a blind approach to this block, the location of the kidneys is an easy location to palpate. Additionally, with this study being performed in a teaching lab there would be ample time between block and beginning surgery due to surgeon inexperience. The technique could further be refined by investigating additional local anesthetic agents with variable onset and duration.

The postoperative pain score results suggest that the use of

ultrasound-guided intraperitoneal block with bupivacaine may not provide any significant advantage over postoperative pain management with non-steroidal anti-inflammatory drugs. This is consistent with the findings of other studies on postoperative pain scores with intraperitoneal bupivacaine (Benito et al., 2016; Benito et al., 2018). The lack of additional intraoperative pain control in the presence of effective postoperative analgesia as found in other studies could indicate that the anti-inflammatory properties of bupivacaine are important for the analgesic effect seen (Cassuto et al., 2006). Despite an anesthetic protocol that included ketamine, alpha-2 agonist, and opioid, this study had a relatively high rescue analgesic rate in both groups (US-IPLA = 42 %, US-S = 30 %) compared to previous studies such as those by Benito et al. (NSAID = 13 %, Bupivacaine = 26 %) which could be due to greater tissue trauma owed to the inexperience of the surgeons in this study. NSAID treatment and intraperitoneal bupivacaine together may have an additive effect, and be beneficial, particularly in scenarios of increased tissue trauma, although this requires further investigation.

This study suggests that the use of ultrasound-guided intraperitoneal block with bupivacaine may not offer significant analgesic benefits in cats undergoing ovariohysterectomy and additionally bupivacaine may have shorter duration of action compared to NSAID treatment. However, there may be an additive benefit to both therapies in conjunction and further studies are necessary to determine if variation of local anesthetic dose, concentration, or volume would improve efficacy, if use of this technique in a multimodal analgesic strategy is beneficial, or if this technique may be beneficial in other surgical procedures.

Several limitations to this study should be considered. First, the study was conducted in a surgery teaching laboratory, which may not fully represent a general clinical setting. The use of veterinary students to perform the surgery and anesthesia may have also introduced variability in the execution of the procedures. However, it is expected that the inexperienced tissue handling would increase nociceptive input during surgery and increase the magnitude of the difference if one existed. Second, the generalized administration into the peritoneal cavity may result in dilution of the bupivacaine to concentrations that are not effective for full nerve blockade. The minimum effective concentration for sensory block in 90 % of patients (MEC90) ranges between 0.24 and

0.27 % depending on block location and the specific study (Takeda et al., 2015; Moura et al., 2016). Our concentration was 0.25 % which is expected to fall in this range, however, small amounts of peritoneal fluid could dilute this beyond an effective concentration. Additional studies in humans show that extremely low concentrations of bupivacaine irrigation (0.02 %) provide postoperative analgesia for laparoscopic cholecystectomy (Jain et al., 2018). This may provide evidence for an anti-inflammatory mechanism of analgesia. Third, the study only evaluated the short-term analgesic effects of bupivacaine, and long-term outcomes (such as chronic pain) were not assessed. Fourth, the study utilized blood pressure management strategies (such as ephedrine and atropine) that may influence HR and SAP which were used as outcome measures. However, with atropine being more frequent in the control group it would be expected that this finding would increase a heart rate disparity due to nociception if there were one. Fifth, isoflurane vaporizer setting was used in this study as opposed to end-tidal isoflurane values. This limitation was mitigated by the use of non-rebreathing systems. Additionally, the study was not powered to detect a small difference in isoflurane vaporizer setting. Lastly, the study only evaluated cats undergoing OHE, and the results may not be applicable to other surgical procedures in the abdomen or species. Further studies evaluating the efficacy of bupivacaine IPLA in other surgical procedures and species are needed, in addition to studies addressing long-term outcomes. The mechanism of postoperative pain control for this technique also warrants further investigation.

5. Conclusions

Preoperative administration of bupivacaine via ultrasound-guided IPLA did not mitigate the hemodynamic response to surgical stimulation, nor did it provide better analgesia than administering robenacoxib at recovery. This study highlights the importance of assessing the efficacy of local anesthetic techniques for surgical pain management in cats and provides insight into the use of IPLA as an analgesic technique for OHE. Future studies to evaluate the timing and dosing of IPLA are necessary to determine if more effective visceral analgesia can be obtained. Ultimately, this study underscores the importance of multimodal analgesia to provide comprehensive pain management for cats undergoing OHE.

Author's contributions

BML: study design, treatment administration, data collection (intraoperative variables, postoperative pain scoring), data management and interpretation, manuscript preparation. AGPG: study design, data collection (intraoperative variables), data interpretation, statistical analysis, manuscript preparation. CCT: study design, data collection (intraoperative variables), manuscript preparation. EWH: study design, treatment preparation, data interpretation, manuscript preparation. LI: data collection (intraoperative variables), manuscript preparation. BW: data collection (intraoperative variables), manuscript preparation. ALH: data collection (intraoperative variables), manuscript preparation. All authors edited the manuscript and approved the final version of the manuscript.

CRedit authorship contribution statement

Brandon M. Lawler: Writing – review & editing, Writing – original draft, Project administration, Investigation, Data curation. **Alonso G.P. Guedes:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Caitlin C. Tearney:** Writing – review & editing, Supervision, Methodology, Investigation, Data curation, Conceptualization. **Lauren Ienello:** Writing – review & editing, Data curation.

Brian Walters: Writing – review & editing, Data curation. **Adrienne L. Haley:** Writing – review & editing, Data curation. **Erin Wendt-Hornick:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare no conflict of interest.

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